
MAKERERE UNIVERSITY
Department of Computer Science
CSC 1105: Mathematics for Computer Science

ASSIGNMENT 2
Polynomial Approximation and Interpolation
Taylor Series and Lagrange Interpolation

Instructions

1. This assignment is due on **September 18, 2026**.
2. Students should work in groups of a maximum of **six (6) members**.
3. The assignment must be **typed, printed, and submitted in hard copy**.
4. Attempt all questions.
5. Show all essential mathematical steps. Answers without sufficient justification may receive little or no credit.
6. Where a particular theorem, approximation technique, or interpolation formula is used, clearly state it.
7. Groups will be required to present and defend their mathematical model in class or online.
8. Where appropriate, clearly explain the practical meaning of your mathematical results.

Modelling the Route of an Underground Water Pipeline

A city authority in an African city is undertaking major road rehabilitation works. Over the years, several underground water pipes have been accidentally cut by road construction equipment because contractors do not have a reliable continuous map showing the exact route followed by the buried pipelines.

The authority possesses some old survey records containing coordinates of selected points through which one of the major water pipelines is known to pass. However, between these known points, the exact mathematical equation of the pipeline route is unavailable.

You have been hired as a mathematics and computing consultant to develop a **polynomial model** that can be incorporated into a simple computer program or Geographic

Information System (GIS). The model should allow engineers to estimate the position of the pipeline between surveyed points before excavation begins.

In this problem, you will investigate two important methods of polynomial approximation:

- (a) Taylor/Maclaurin series, when the mathematical function is known; and
- (b) Lagrange interpolation, when only measured coordinate data are known.

Part A: Taylor Series as a Polynomial Approximation

Before reconstructing the horizontal route of the pipeline, the engineering team considers a short section for which the approximate depth of the pipe is already described by the function

$$D(x) = 1.8e^{-0.1x},$$

where

- x is the distance along the road measured in units of 100 metres, and
- $D(x)$ is the approximate depth of the pipe below the road surface, measured in metres.

Computers can evaluate polynomials very efficiently. The engineering team therefore wants to replace the exponential function locally by a polynomial.

Recall that if $f(x)$ is sufficiently differentiable, its Taylor series about $x = a$ is

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f'''(a)}{3!}(x - a)^3 + \cdots,$$

or, using summation notation,

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x - a)^k$$

When $a = 0$, the expression becomes the Maclaurin series

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k$$

A1. Find

$$D'(x), \quad D''(x), \quad D'''(x), \quad D^{(4)}(x).$$

A2. Evaluate

$$D(0), \quad D'(0), \quad D''(0), \quad D'''(0), \quad D^{(4)}(0).$$

A3. Hence construct the **fourth-degree Maclaurin polynomial** $T_4(x)$ for $D(x)$.

Your final answer should have the form

$$T_4(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4.$$

A4. Use your polynomial $T_4(x)$ to estimate the depth of the pipeline at

$$x = 1.5.$$

State the corresponding physical distance along the road in metres.

A5. Calculate the actual value

$$D(1.5) = 1.8e^{-0.15}$$

using a calculator and determine the absolute approximation error

$$E = |D(1.5) - T_4(1.5)|.$$

A6. Briefly explain why Taylor series can be used in this part of the problem but cannot immediately be used to reconstruct the unknown horizontal route of the pipeline from a table of coordinates alone.

Part B: Recovering the Pipeline Route from Surveyed Coordinates

For another section of the road, there is no known equation describing the horizontal path of the pipeline. The authority only has the following surveyed points.

Let

- x denote the distance along the road, measured in units of 100 metres from a fixed reference point; and
- y denote the perpendicular position of the pipe, in metres, relative to a fixed survey reference line.

The available coordinates are:

Survey Point	x	y
P_0	0	5
P_1	2	13
P_2	5	40
P_3	7	68

Notice that the x -coordinates are **not equally spaced**. The authority therefore decides to use Lagrange interpolation.

How the Lagrange Polynomial is Constructed

Suppose that we are given $n + 1$ data points

$$(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n).$$

We want to construct a polynomial $P_n(x)$ satisfying

$$P_n(x_i) = y_i, \quad i = 0, 1, \dots, n.$$

For each data point, define a Lagrange basis polynomial $L_j(x)$ such that

$$L_j(x_j) = 1$$

but

$$L_j(x_i) = 0, \quad i \neq j.$$

This is achieved by defining

$$L_j(x) = \prod_{\substack{i=0 \\ i \neq j}}^n \frac{x - x_i}{x_j - x_i}$$

The interpolating polynomial is then obtained by multiplying each basis polynomial by its corresponding y -coordinate and adding:

$$P_n(x) = \sum_{j=0}^n y_j L_j(x)$$

Therefore,

$$P_n(x) = \sum_{j=0}^n y_j \prod_{\substack{i=0 \\ i \neq j}}^n \frac{x - x_i}{x_j - x_i}$$

For four data points, the polynomial is

$$P_3(x) = y_0 L_0(x) + y_1 L_1(x) + y_2 L_2(x) + y_3 L_3(x).$$

Explicitly,

$$\begin{aligned}
P_3(x) = & y_0 \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} \\
& + y_1 \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} \\
& + y_2 \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} \\
& + y_3 \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)}.
\end{aligned}$$

Part C: Constructing the Pipeline Model

Using the surveyed pipeline coordinates given above, answer the following questions.

C1. Identify

$$x_0, x_1, x_2, x_3$$

and

$$y_0, y_1, y_2, y_3.$$

C2. Construct the four Lagrange basis polynomials

$$L_0(x), \quad L_1(x), \quad L_2(x), \quad L_3(x).$$

C3. Hence construct

$$P_3(x) = y_0 L_0(x) + y_1 L_1(x) + y_2 L_2(x) + y_3 L_3(x).$$

C4. Expand and simplify your result until the pipeline model is written in standard polynomial form

$$\boxed{P_3(x) = ax^3 + bx^2 + cx + d}.$$

If any coefficient becomes zero during simplification, clearly indicate this.

C5. Verify your polynomial by calculating

$$P_3(0), \quad P_3(2), \quad P_3(5), \quad P_3(7).$$

Explain why these values provide evidence that your interpolation is correct.

Part D: Using the Model During Road Construction

The road contractor plans to excavate at a position corresponding to

$$x = 3.5.$$

D1. Convert $x = 3.5$ into the actual distance in metres from the reference point.

D2. Use your Lagrange polynomial to estimate the perpendicular position of the water pipeline at

$$x = 3.5.$$

D3. An excavator is expected to operate at the point

$$(3.5, 25.0).$$

Based on your polynomial estimate, determine the perpendicular distance between the expected pipeline position and the planned excavation point.

D4. If the authority requires a minimum horizontal safety clearance of 3 metres from the estimated pipeline position, determine whether the excavation should proceed at the proposed position.

Justify your conclusion mathematically.

Part E: Rate of Change of the Pipeline Route

Once the polynomial $P_3(x)$ has been obtained, information about the direction of the pipeline can be obtained by differentiation.

E1. Differentiate your polynomial to obtain

$$P'_3(x).$$

E2. Calculate

$$P'_3(3.5).$$

E3. Explain, in the context of the pipeline, what $P'_3(3.5)$ tells an engineer about the route of the pipe near that location.

E4. Determine whether the pipeline is locally increasing or decreasing relative to the survey reference line at $x = 3.5$. Give a mathematical reason for your answer.

Part F: Comparing Taylor and Lagrange Approximation

Write a short discussion of approximately **150–250 words** comparing Taylor series and Lagrange interpolation.

Your discussion should address the following points:

- (i) What information must be known before Taylor series can be used?
- (ii) What information must be known before Lagrange interpolation can be used?
- (iii) Why is Lagrange interpolation particularly suitable for the authority's historical pipeline coordinate data?
- (iv) What is the difference between **approximating a known function** using a Taylor polynomial and **constructing a function from known data points** using Lagrange interpolation?
- (v) State at least two practical limitations or risks that the city authority should consider before using the polynomial model for actual excavation work.

Part G: Python Implementation of Lagrange Interpolation

The city authority would like to incorporate the mathematical model into a computer program that can be used by engineers to estimate the position of the pipeline at locations where no direct survey measurement is available.

You are required to implement the Lagrange interpolation method using **Python**.

Recall that for $n + 1$ data points

$$(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n),$$

the Lagrange interpolating polynomial is

$$P_n(x) = \sum_{j=0}^n y_j L_j(x),$$

where

$$L_j(x) = \prod_{\substack{i=0 \\ i \neq j}}^n \frac{x - x_i}{x_j - x_i}.$$

Hence,

$$P_n(x) = \sum_{j=0}^n y_j \prod_{\substack{i=0 \\ i \neq j}}^n \frac{x - x_i}{x_j - x_i}$$

G1. Write a Python function called

`lagrange_interpolation(x_values, y_values, x)`

which evaluates the Lagrange interpolating polynomial at a specified value of x . Your program should implement the Lagrange formula directly rather than using a ready-made interpolation function from a Python library.

G2. Your function should be able to work with an arbitrary number of coordinate pairs. In particular, it should:

- (i) accept the x -coordinates as a list;
- (ii) accept the corresponding y -coordinates as a list;
- (iii) compute each Lagrange basis polynomial $L_j(x)$;
- (iv) multiply each $L_j(x)$ by the corresponding y_j ;
- (v) add all the terms to obtain $P_n(x)$; and
- (vi) return the estimated value $P_n(x)$.

G3. Test your Python function using the pipeline coordinates

$$(0, 5), \quad (2, 13), \quad (5, 40), \quad (7, 68).$$

Therefore, your Python program should contain data equivalent to

$$x_{\text{values}} = [0, 2, 5, 7]$$

and

$$y_{\text{values}} = [5, 13, 40, 68].$$

G4. Use your Python function to estimate the position of the pipeline at

$$x = 3.5.$$

Display the result clearly.

G5. Compare the value obtained from your Python program with the value that you calculated manually in Part D.

Calculate the difference

$$|P_{\text{manual}}(3.5) - P_{\text{Python}}(3.5)|.$$

Explain what you expect this difference to be and why.

G6. Test the program at all the original survey coordinates by evaluating

$$P_3(0), \quad P_3(2), \quad P_3(5), \quad P_3(7).$$

Present the results in a table similar to the following:

x	Original Survey Value	Python Output
0	5	
2	13	
5	40	
7	68	

Explain why a correctly implemented Lagrange interpolating polynomial should reproduce the original survey values at these points.

G7. Use your Python function to estimate the pipeline position at each of the following unsurveyed locations:

$$x = 1, \quad x = 3, \quad x = 4, \quad x = 6.$$

Present your results in a table.

G8. Modify your program so that it generates estimated pipeline coordinates at intervals of 0.5 for

$$0 \leq x \leq 7.$$

That is, calculate

$$P_3(0), P_3(0.5), P_3(1.0), P_3(1.5), \dots, P_3(7.0).$$

G9. Using Python and the `matplotlib` library, produce a graph containing:

- (i) the original surveyed pipeline coordinates as points; and
- (ii) the interpolated polynomial curve obtained from your program.

Label the horizontal and vertical axes appropriately and give the graph a suitable title.

G10. From the graph and the computed values, briefly discuss whether the polynomial appears to provide a reasonable representation of the pipeline route between the surveyed points.

Submission Requirements for the Python Component

For this part of the assignment, each group must include:

- (i) the complete Python source code;
- (ii) the program output for all required test values;
- (iii) the table comparing the original coordinates with the Python results;
- (iv) the generated graph showing the survey points and interpolating polynomial; and
- (v) a short explanation of how the Python implementation corresponds to the mathematical Lagrange formula.

Note: You should implement the Lagrange interpolation formula. Functions such as `scipy.interpolate.lagrange`, `numpy.polyfit`, or other functions that automatically construct the interpolating polynomial should not be used for the main implementation. They may only be used afterwards, if desired, to verify the result obtained from your own implementation.

Part H: Extension

Suppose that, after further surveying, the authority discovers an additional pipeline coordinate

$$(x_4, y_4) = (8, 85).$$

Without carrying out the entire calculation, answer the following:

- H1.** What would be the maximum degree of the Lagrange interpolating polynomial constructed from all five points?
- H2.** Write the general form of the new interpolation model.
- H3.** Write the expression for the new basis polynomial $L_4(x)$.
- H4.** Suggest one way in which the authority could improve the reliability of the mathematical model before using it for excavation decisions.

End